STEP 01 OF 15 NETWORK FOUNDATION

Create the VPC and Subnets

Draw the map before you build the city

WHY THIS STEP?

Everything else lives inside this private network, so it is built first. The VPC and its six subnets decide which resources can ever talk to each other.

Sign in to the AWS Console, choose **Canada (Central)** in the Region selector and keep it there for the whole project. Use the console search bar to open each service.

Create the VPC

1. Search for **VPC** and open **Your VPCs > Create VPC**.
2. Choose **VPC only** and set the name tag to `Three-Tier-Project-VPC`.
3. Select **IPv4 CIDR manual input** and enter `10.0.0.0/16`. Keep **No IPv6 CIDR block** and tenancy **Default**.
4. Click **Create VPC**.

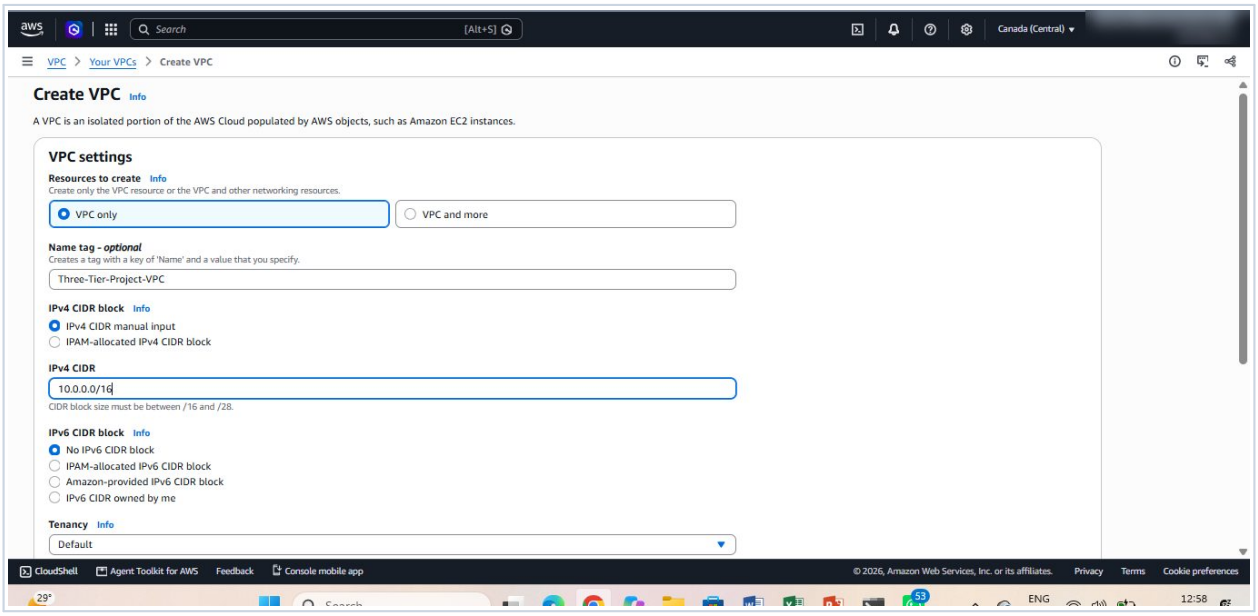


Figure 1. Create VPC: VPC only, name tag Three-Tier-Project-VPC, IPv4 CIDR 10.0.0.0/16.

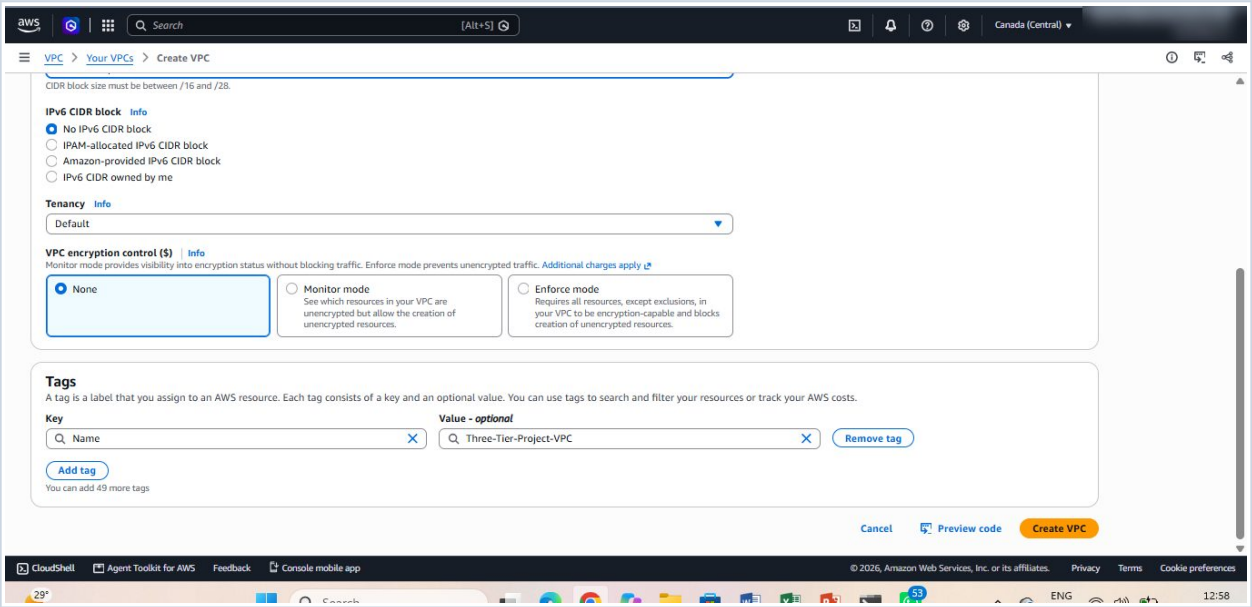


Figure 2. Scroll down, confirm the Name tag and click Create VPC.

Create the subnets

Create six subnets in the new VPC, three per Availability Zone (ca-central-1a and ca-central-1b). Go to **VPC > Subnets > Create subnet**, choose **Three-Tier-Project-VPC** and enter each subnet as below.

Subnet name	Availability Zone	Type	IPv4 CIDR block
Web-Tier-Subnet-1-Public	ca-central-1a	Public	10.0.1.0/24
App-Tier-Subnet-1-Private	ca-central-1a	Private	10.0.0.0/24
DB-Tier-Subnet-1-Private	ca-central-1a	Private	10.0.3.0/24
Web-Tier-Subnet-2-Public	ca-central-1b	Public	10.0.4.0/24
App-Tier-Subnet-2-Private	ca-central-1b	Private	10.0.5.0/24
DB-Tier-Subnet-2-Private	ca-central-1b	Private	10.0.6.0/24

PRO TIP

I created the first four subnets one at a time, then used **Add new subnet** to enter the last two on a single page. Either way works.

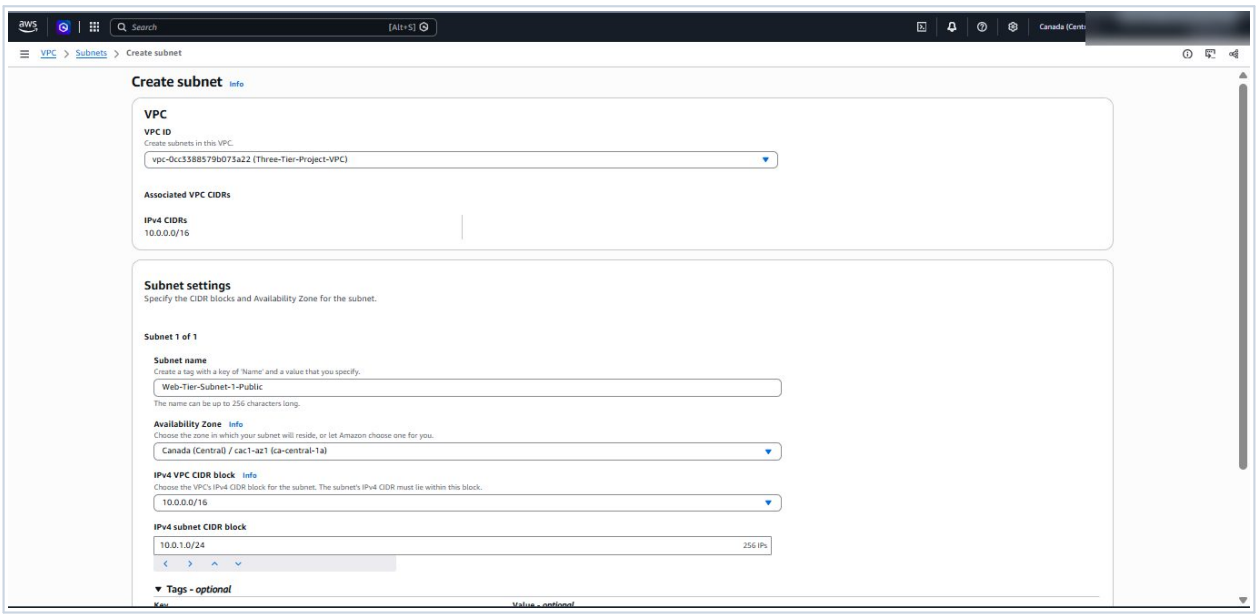


Figure 3. Subnet 1: Web-Tier-Subnet-1-Public in ca-central-1a (10.0.1.0/24).

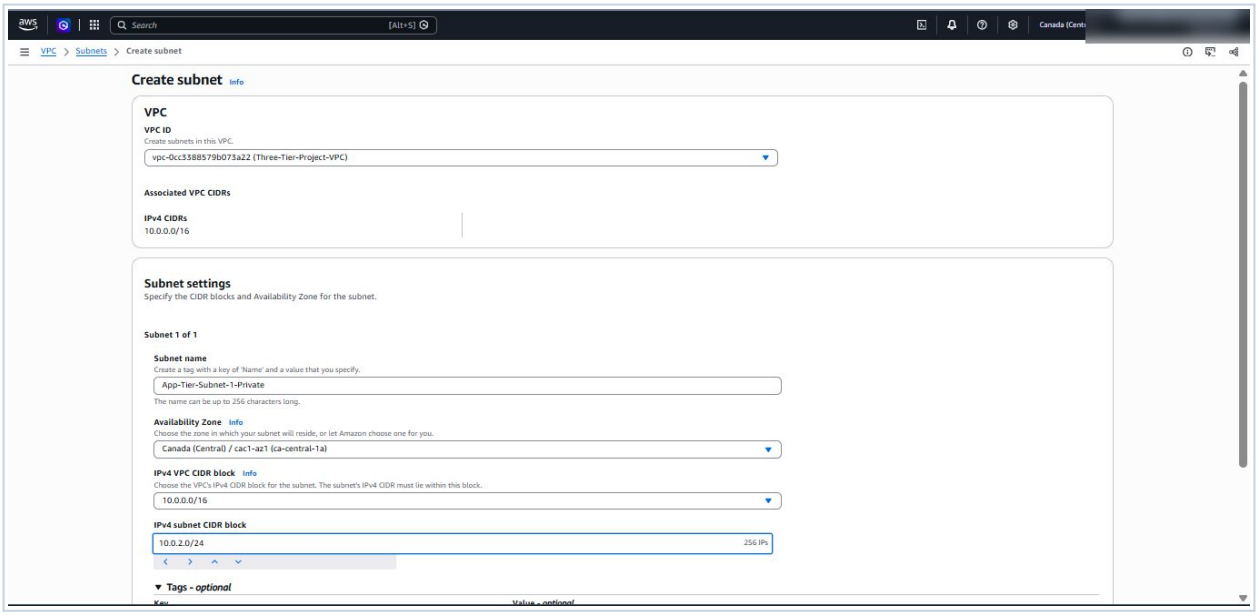


Figure 4. Subnet 2: App-Tier-Subnet-1-Private in ca-central-1a (10.0.0.0/24).

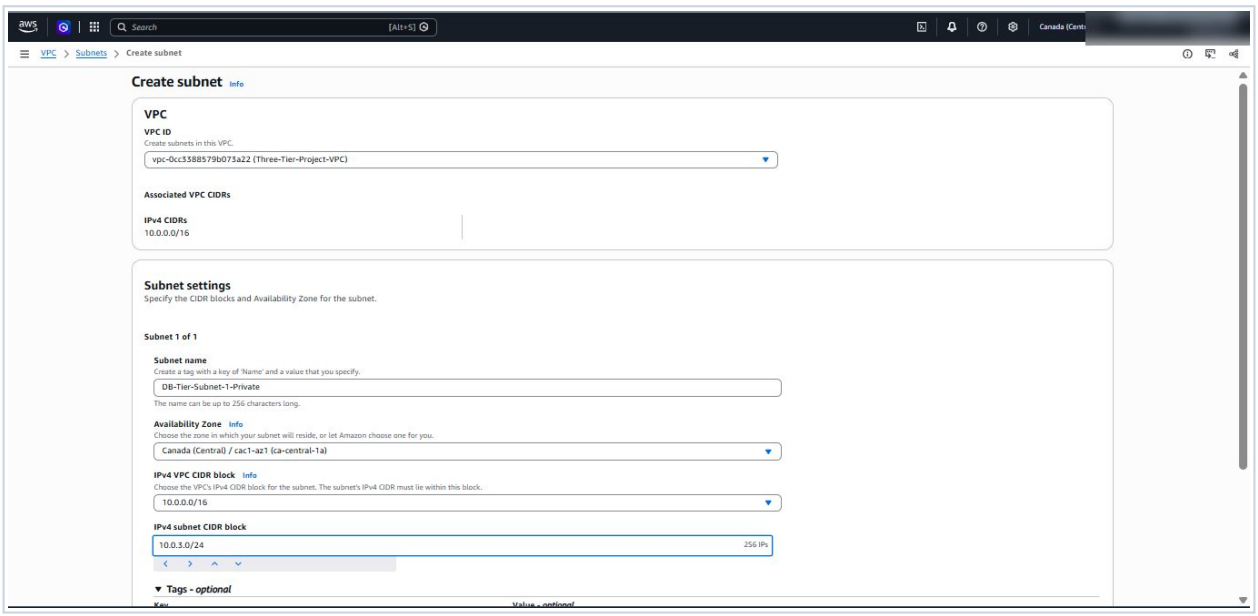


Figure 5. Subnet 3: DB-Tier-Subnet-1-Private in ca-central-1a (10.0.3.0/24).

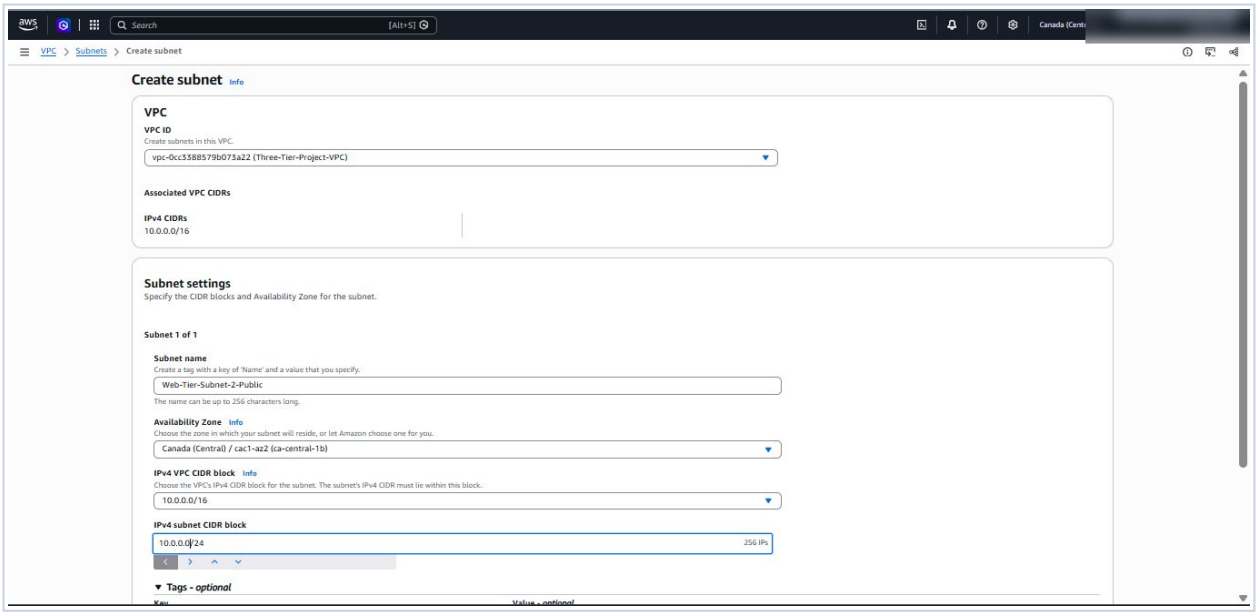


Figure 6. Subnet 4: Web-Tier-Subnet-2-Public in ca-central-1b (10.0.4.0/24).

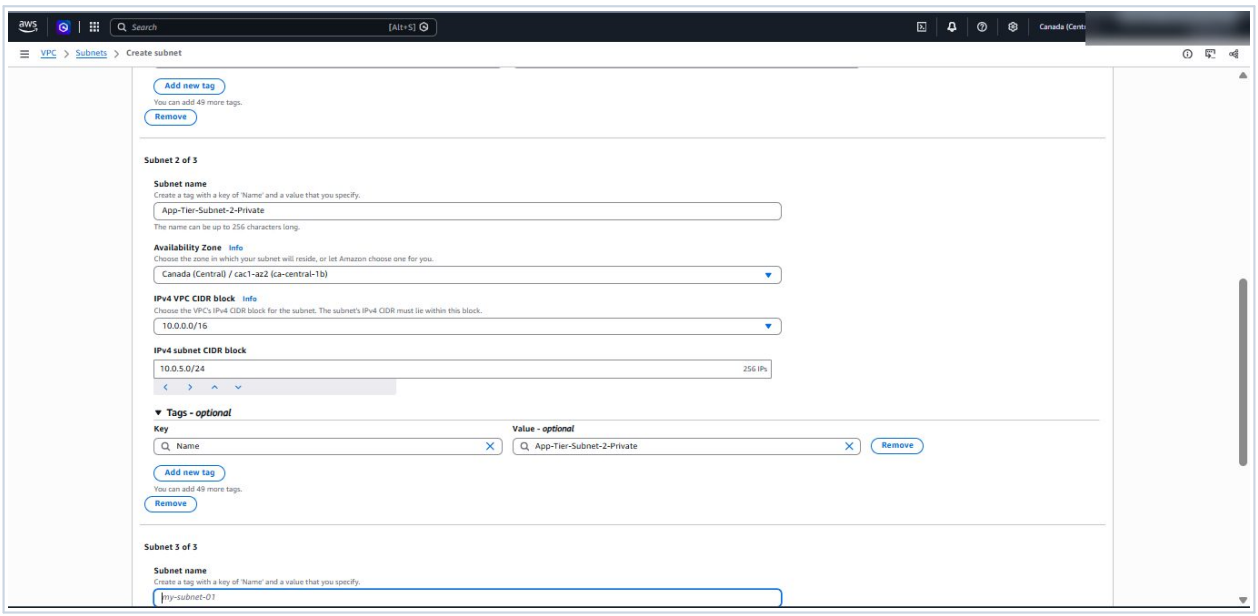


Figure 7. Subnet 5: App-Tier-Subnet-2-Private in ca-central-1b (10.0.5.0/24), added with Add new subnet.

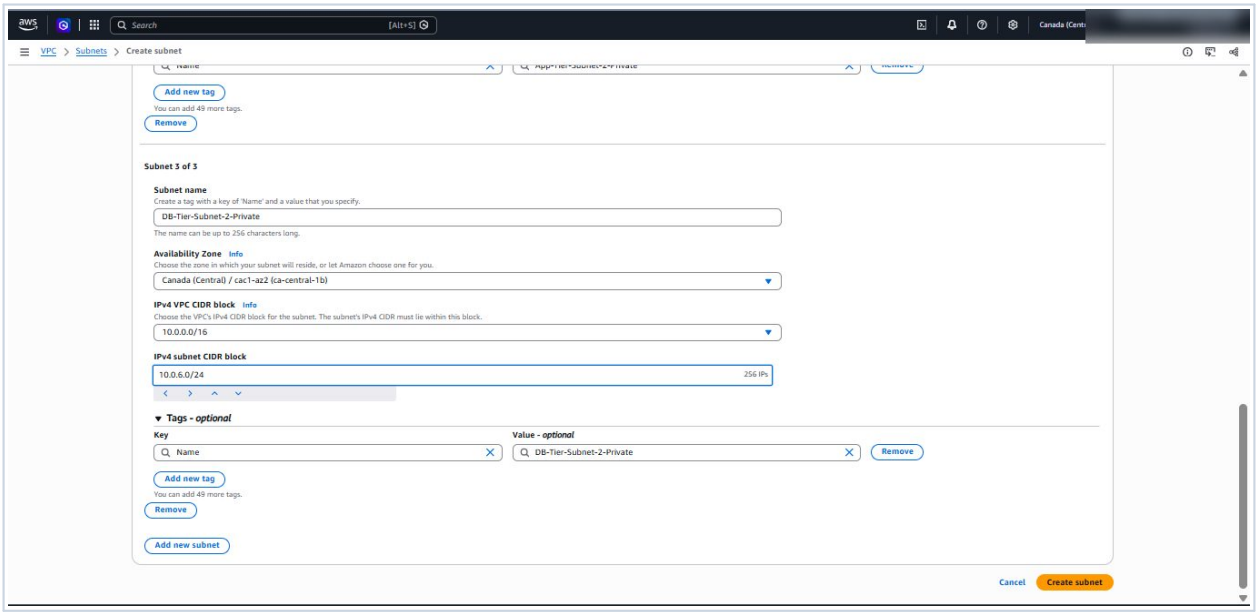


Figure 8. Subnet 6: DB-Tier-Subnet-2-Private in ca-central-1b (10.0.6.0/24). Click Create subnet.

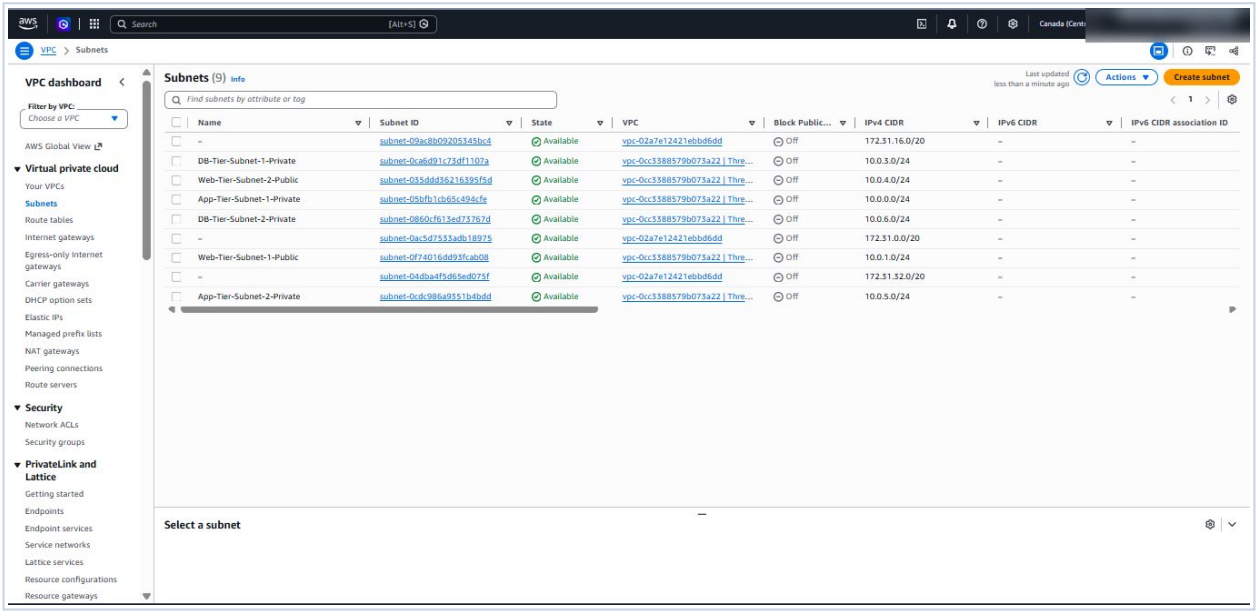


Figure 9. The Subnets list after creation: the six project subnets sit beside the three default-VPC subnets.

Enable auto-assign public IP on the web subnets

1. Select **Web-Tier-Subnet-1-Public** and choose **Actions > Edit subnet settings**.
2. Tick **Enable auto-assign public IPv4 address** and click **Save**.
3. Repeat for **Web-Tier-Subnet-2-Public**.

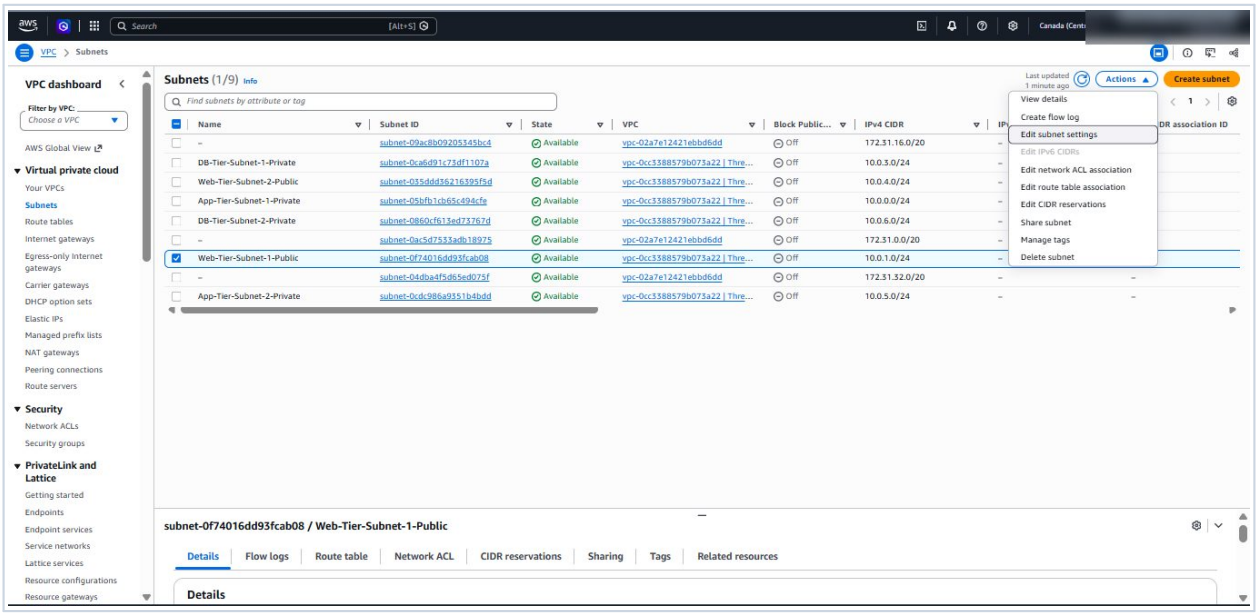


Figure 10. Select Web-Tier-Subnet-1-Public, then Actions > Edit subnet settings.

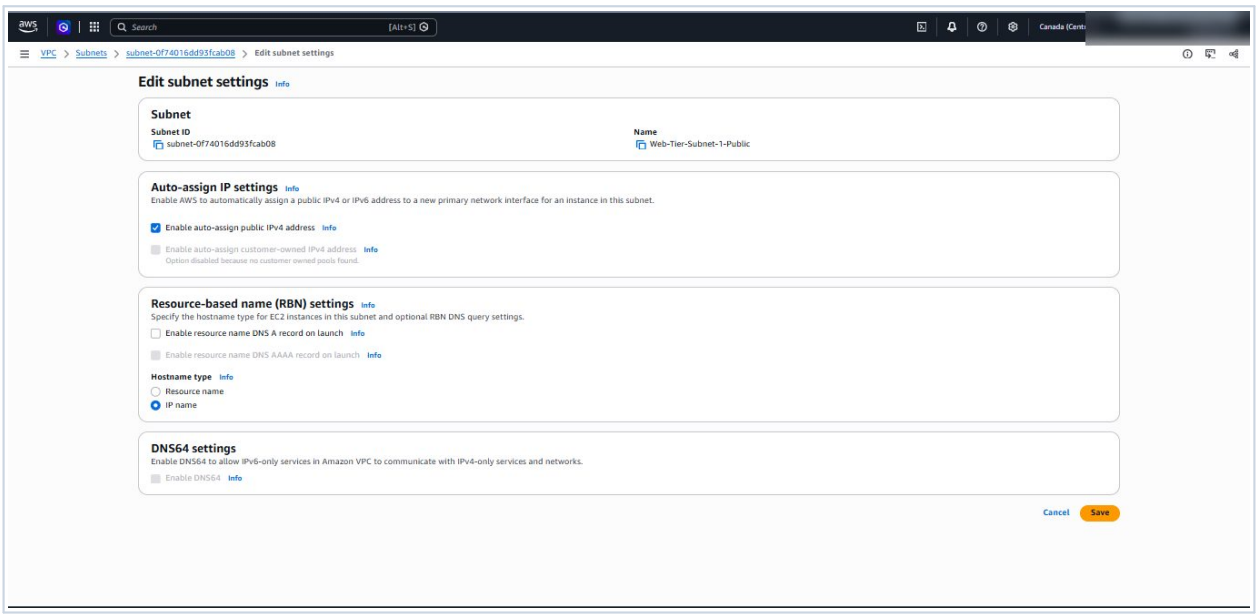


Figure 11. Enable auto-assign public IPv4 address for Web-Tier-Subnet-1-Public and save.

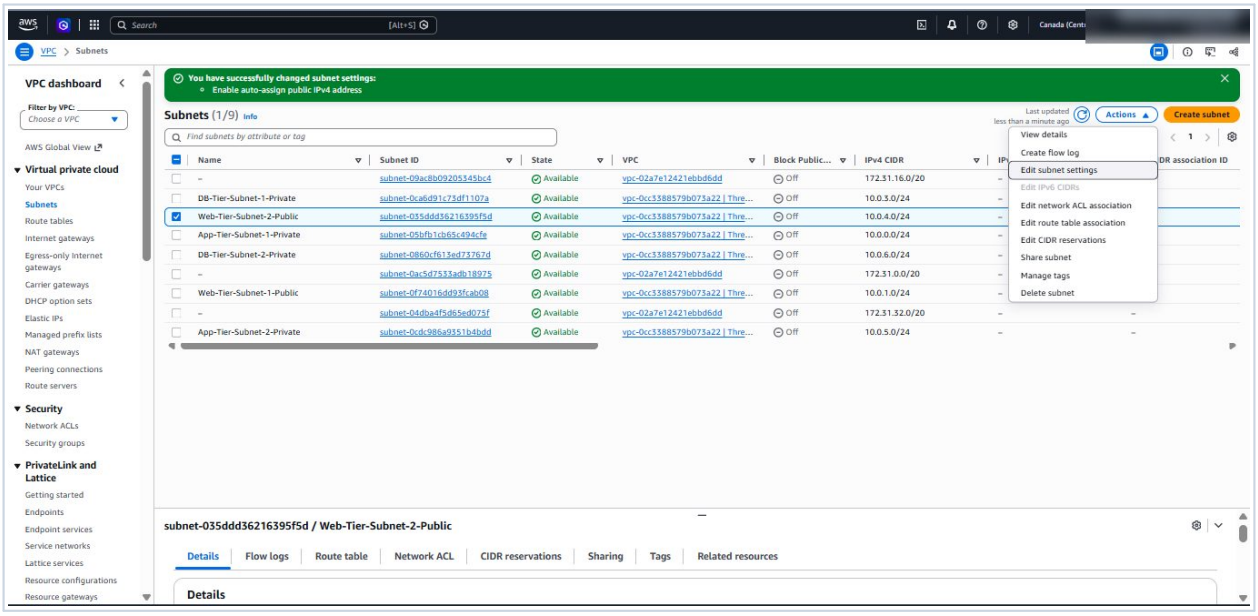


Figure 12. The success banner confirms subnet 1. Now select Web-Tier-Subnet-2-Public and open Edit subnet settings.

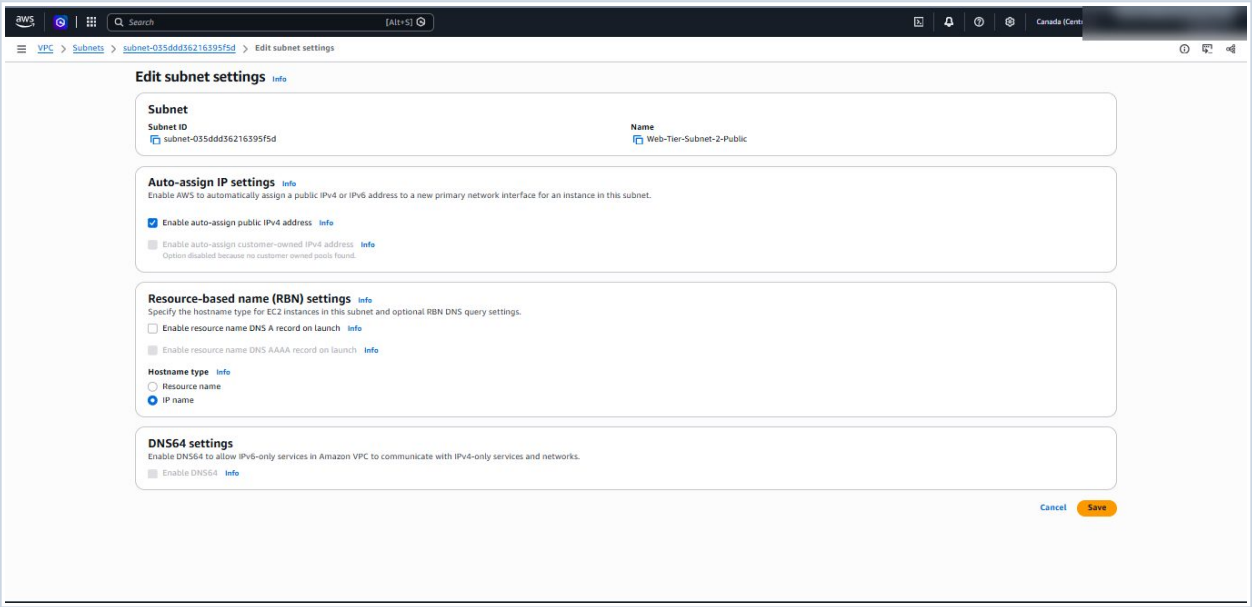


Figure 13. Enable auto-assign public IPv4 address for Web-Tier-Subnet-2-Public and save.

CHECKPOINT

Six subnets exist in Three-Tier-Project-VPC, and both web subnets hand out public IPv4 addresses automatically.